

AP CSP Vocabulary Glossary

Working in Python · alphabetical · 300 terms · all terms

Terms marked ^{AP} appear on the College Board AP CSP exam's own vocabulary list. A superscript chapter number links to where this book defines the term.

Vocabulary drawn from *Working in Python*, a fork of Allen Downey's *Think Python*, 3rd edition (CC BY-NC-SA 4.0), and this course's own AP CSP vocabulary reference. AP CSP definitions here are written in this course's own words, not copied from the College Board's Course and Exam Description.

A

absolute path¹³ A path that does not depend on the current directory.

abstraction^{AP} Reducing complexity by exposing essential features and hiding implementation detail.

accumulator¹⁰ A variable used in a loop to add up or accumulate a result.

aggregation of information^{AP} Combining separately harmless pieces of data to identify or profile a person.

algorithm^{AP} A finite sequence of steps that solves a problem or completes a task. Can be written in English, pseudocode, or code.

algorithmic efficiency^{AP} An estimate of the computational resources an algorithm uses as a function of input size.

aliased⁹ If there is more than one variable that refers to an object, the object is aliased.

analog data^{AP B} Values that vary continuously and smoothly, with no steps.

anonymization^{AP} Removing identifying details from data — and its limits, since aggregation can re-identify.

API^{AP} The specification of how a library's procedures are called and how they behave.

argument^{AP 2} A value provided to a function when the function is called.

arithmetic operator¹ A symbol, like + and *, that denotes an arithmetic operation like addition or multiplication.

ASCII^{AP B} A table assigning a number from 0 to 127 to each of a small set of characters.

assignment statement^{AP 2} A statement that assigns a value to a variable.

attribute^{9,14} A named value associated with an object; called an instance variable when it belongs to a class instance.

authentication^{AP} Verifying that a user is who they claim to be.

B

bandwidth^{AP} Maximum data transmittable per unit time, in bits per second.

base case⁵ A conditional branch in a recursive function that does not make a recursive call.

beneficial and harmful effects^{AP} Every computing innovation has both, and they can fall on different groups of people.

bias^{AP} Systematic unfairness embedded in an algorithm or in the data it was built from.

bigram¹² A sequence of two elements, often words.

binary^{AP B} Base-2 representation, using only the digits 0 and 1.

binary mode¹³ A way of opening a file so the contents are interpreted as a sequence of bytes rather than a sequence of characters.

binary search^{AP} Repeatedly halving a sorted list to locate a value.

bit^{AP B} A single binary digit, 0 or 1.

block⁵ One or more statements indented to indicate they are part of another statement. Statements in a block are frequently said to have the same scope.

body³ The sequence of statements inside a function definition.

boolean expression^{AP 5} An expression whose value is either True or False.

boundary case^A A test case at the value where a function's behavior changes, such as zero, an empty string, or the first or last item.

branch⁵ One of the alternative sequences of statements in a conditional statement.

bug¹ An error in a program.

byte^{AP B} Eight bits. Enough to hold one of 256 values.

C

call graph¹⁰ A diagram that shows every frame created during the execution of a program, with an arrow from each caller to each callee.

canvas⁴ A window used to display graphical elements including lines, circles, rectangles, and other shapes.

certificate authority^{AP} An organization that issues and vouches for digital certificates.

chained conditional⁵ A conditional statement with a series of alternative branches.

character⁸ An element of a string, including letters, numbers, and symbols.

character encoding^B The agreement about which numbers stand for which characters. Read bits with the wrong encoding and you get the right data as the wrong text.

child class¹⁷ A class that inherits from another class.

citizen science^{AP} Scientific research carried out partly by volunteers using their own devices.

class¹⁴ A programmer-defined type. A class definition creates a new class object.

class object¹⁴ An object that represents a class — it is the result of a class definition.

class variable¹⁷ A variable defined inside a class definition, but not inside any method.

code segment^{AP} A portion of a program, one or more lines, treated as a unit.

collaboration^{AP} Working with others to develop a computing innovation; benefits from varied perspectives and skills.

comment^{AP 2} Text included in a program that provides information about the program but has no effect on its execution.

compound conditional^{AP} A single condition combining multiple Boolean expressions with AND, OR, or NOT.

compression ratio^B Compressed size divided by original size.

computer network^{AP} Interconnected computing devices able to send and receive data.

computer virus^{AP} Malware that copies itself, typically by attaching to other programs or files.

computing device^{AP} A physical artifact that can run a program.

computing innovation^{AP} A product, service, or concept that includes a program as an integral part of its function. May be physical, software, or conceptual.

computing system^{AP} A group of computing devices and programs working together.

concatenation^{AP 1} Joining two strings end-to-end.

condition⁵ The boolean expression in a conditional statement that determines which branch runs.

conditional expression¹⁸ An expression that uses a conditional to select one of two values.

conditional statement^{AP 5} A statement that controls the flow of execution depending on some condition. Informally, this is usually an if statement, which might include an elif and an else. (*AP calls this selection.*)

configuration data¹³ Data, often stored in a file, that specifies what a program should do and how.

constant^{AP} A named value that does not change during execution.

cookies^{AP} Small data files stored by a site to track a user's state or behavior.

copyright^{AP} The exclusive right of a creator to reproduce, distribute, and display a work.

correlation^{AP} An association between two variables where changes in one accompany changes in the other.

counter⁷ A variable used to count something, usually initialized to zero and then incremented.

creative commons^{AP} A set of public licenses letting a creator pre-authorize specific reuse under stated conditions.

crowdsourcing^{AP} Gathering input, funding, or labor from many people over the Internet.

current working directory¹³ The default directory used by a program unless another directory is specified.

D

data^{AP} Values that can be stored and processed by a program.

data abstraction^{AP} Using a data structure, typically a list, to represent related values as one named unit, hiding the individual pieces.

data cleaning^{AP} Making data uniform and consistent without changing its meaning.

data mining^{AP} Analyzing large data sets to find patterns and relationships.

data set^{AP} A collection of related data organized for analysis.

data structure¹¹ A collection of values, organized to perform certain operations efficiently.

database¹³ A file whose contents are organized to perform certain operations efficiently.

dead code⁶ Part of a program that can never run, often because it appears after a return statement.

debugging^{AP 1} The process of finding and correcting errors.

decidable problem^{AP} A yes/no problem for which an algorithm can always produce the correct answer.

decimal^{AP B} Base-10, the system you already use.

decrement⁷ Decrease the value of a variable.

deep copy¹⁶ A copy operation that also copies nested objects.

default value¹² The value assigned to a parameter if no argument is provided.

delegation¹⁷ When one method passes responsibility to another method to do most or all of the work.

delimiter⁹ A character or string used to indicate where a string should be split.

deserialization¹³ Converting a string to an object.

design-first development¹⁴ A way of developing programs with more careful planning than prototype and patch.

deterministic¹² A deterministic program does the same thing each time it runs, given the same inputs.

development plan⁴ A process for writing programs.

dictionary¹⁰ An object that contains key-value pairs, also called items.

digest¹³ The result of a hash function, especially when it is used to check whether two objects are the same.

digital certificate^{AP} A credential issued by an authority binding a public key to an identity.

digital data^{AP B} Values represented in discrete steps, ultimately as bits.

digital divide^{AP} Unequal access to computing and the Internet across socioeconomic, geographic, or demographic lines.

directory¹³ A collection of files and other directories.

distributed computing^{AP} Multiple devices in different locations running parts of one program.

DNS^{AP} Translates domain names into IP addresses.

docstring^{AP 4.A} A string at the beginning of a function that documents what the function does; unlike a comment, it is stored on the function and can be read by the program. (*AP calls this program documentation.*)

doctest^A An example call and its expected result, written inside a docstring, that can be run automatically to check the function.

dot operator² The operator, `.`, used to access a function in another module by specifying the module name followed by a dot and the function name.

E

edge case^A A test case at an unusual or extreme input, where a function is most likely to be wrong.

element^{AP 9} One of the values in a list or other sequence.

empty string⁸ A string that contains no characters and has length 0.

encapsulation⁴ The process of transforming a sequence of statements into a function definition.

encode¹⁷ To represent one set of values using another set of values by constructing a mapping between them.

encryption^{AP} Encoding data so only holders of the key can read it.

enumerate object¹¹ The result of calling the built-in function `enumerate`, can be used to loop through a sequence of tuples.

ephemeral¹³ An ephemeral program typically runs for a short time and, when it ends, its data are lost.

equivalent⁹ Having the same value.

evaluate² Perform the operations in an expression in order to compute a value.

event^{AP} An action supplied to a program as input: key press, click, sensor reading.

event-driven programming^{AP} Statements execute in response to events rather than in top-to-bottom sequence.

exception² An error that is detected while the program is running.

execute² Run a statement and do what it says.

expected value^A What a test says the answer should be, as opposed to what the code actually produced.

expression^{AP 1} A combination of variables, values, and operators.

F

f-string¹³ A string that has the letter `f` before the opening quotation mark, and contains one or more expressions in curly braces.

factory¹⁸ A function used to create objects, often passed as a parameter to a function.

fail^{7.A} A test whose actual result does not match its expected value.

fault tolerance^{AP} Continuing to operate when components fail.

file object⁷ An object that represents an open file and keeps track of which parts of the file have been read or written.

filtering^{AP 10} Looping through a sequence and selecting or omitting elements. (*AP calls this data filtering.*)

floating-point¹ A type that represents integers and numbers with decimal parts.

formal language¹ Any of the languages that people have designed for specific purposes, such as representing mathematical ideas or computer programs. All programming languages are formal languages.

format specifier¹⁴ In an f-string, a format specifier determines how a value is converted to a string.

frame³ A box in a stack diagram that represents a function call. It contains the local variables and parameters of the function.

function^{AP 1} A named sequence of statements that performs some useful operation. Functions may or may not take arguments and may or may not produce a result. (*AP calls this a procedure.*)

function call¹ An expression — or part of an expression — that runs a function. It consists of the function name followed by an argument list in parentheses.

function definition³ A statement that creates a function.

function object³ A value created by a function definition. The name of the function is a variable that refers to a function object.

functional programming style¹⁴ A way of programming that uses pure functions whenever possible.

G

generalization⁴ The process of replacing something unnecessarily specific (like a number) with something appropriately general (like a variable or parameter).

generator expression¹⁸ Similar to a list comprehension except that it does not create a list.

H

hand tracing^{AP A} Working through code on paper line by line, writing down each variable's value, in order to find an error without running the program.

hash function^{10,13} A function that takes an object and computes an integer — sometimes called a digest — used to locate a key in a hash table or to check whether two objects are the same.

hash table¹⁰ A collection of key-value pairs organized so that we can look up a key and find its value efficiently.

hashable¹⁰ Immutable types like integers, floats and strings are hashable. Mutable types like lists and dictionaries are not.

header³ The first line of a function definition.

heuristic^{AP} An approach that finds a good-enough solution when finding the optimal one is impractical.

hexadecimal^{AP B} Base-16, using 0 through 9 and A through F. Four bits per digit, so one byte is exactly two hex digits.

HTTP / HTTPS^{AP} The Web's request/response protocol; HTTPS adds encryption in transit.

I

identical⁹ Being the same object (which implies equivalence).

immutable⁸ If the elements of an object cannot be changed, the object is immutable.

import statement² A statement that reads a module file so we can use the variables and functions it contains.

increment⁷ Increase the value of a variable.

incremental development^{AP 6} A program development plan intended to avoid debugging by adding and testing only a small amount of code at a time.

index^{AP 8} An integer value used to select an item in a sequence, such as a character in a string. In Python indices start from 0.

infinite loop^{AP} A loop whose ending condition never becomes true.

infinite recursion⁵ A recursion that doesn't have a base case, or never reaches it. Eventually, an infinite recursion causes a runtime error.

information^{AP} Meaning, patterns, or insight extracted from data.

inheritance¹⁷ The ability to define a new class that is a modified version of a previously defined class.

initialize⁷ Create a new variable and give it a value.

input validation⁶ Checking the parameters of a function to make sure they have the correct types and values.

instance¹⁴ An object that belongs to a class.

instance method¹⁵ A method that must be invoked with an object as receiver.

instantiation¹⁴ The process of creating an object that belongs to a class.

integer^{AP 1} A type that represents numbers with no fractional or decimal part.

integer division¹ An operator, //, that divides two numbers and rounds down to an integer.

intellectual property^{AP} Creative or technical work legally owned by its creator.

interface design^{AP 4} A process for designing the interface of a function, which includes the parameters it should take. (*AP calls this procedural abstraction.*)

internet^{AP} A network of interconnected networks using standardized open protocols.

invariant¹⁵ A condition that should always be true during the execution of a program.

invocation⁸ An expression — or part of an expression — that calls a method.

IP^{AP} Protocol for addressing devices and routing packets between them.

IP address^{AP} The unique identifier assigned to a device on a network.

item¹⁰ In a dictionary, another name for a key-value pair.

iterative development^{AP} Repeated cycles of build, feedback, and revision; earlier phases get revisited.

K

key¹⁰ An object that appears in a dictionary as the first part of a key-value pair.

key-value stores¹³ A database whose contents are organized like a dictionary with keys that correspond to values.

keylogging^{AP} Recording keystrokes to capture passwords and confidential input.

keyword² A special word used to specify the structure of a program.

keyword argument⁴ An argument that includes the name of the parameter.

L

linear search^{AP 7} A computational pattern that searches through a sequence of elements and stops when it finds what it is looking for.

list^{AP 9} An object that contains a sequence of values. (*on the AP exam, list indices start at 1; in Python, at 0.*)

list comprehension¹⁸ A concise way to loop through a sequence and create a list.

local variable³ A variable defined inside a function, and which can only be accessed inside the function.

logical operator^{AP 5} One of the operators that combines boolean expressions, including and, or, and not.

loop^{AP 3} A statement that runs one or more statements, often repeatedly. (*AP calls this iteration.*)

loop variable⁷ A variable defined in the header of a for loop.

lossless compression^{AP B} Reduces size while allowing the original to be reconstructed exactly.

lossy compression^{AP B} Reduces size further, but only an approximation of the original can be recovered.

M

machine learning^{AP} Systems that improve at a task from data rather than from explicit programming.

malware^{AP} Software intended to damage or take control of a system.

mapping¹⁰ A relationship in which each element of one set corresponds to an element of another set.

memo¹⁰ A computed value stored to avoid unnecessary future computation.

metadata^{AP} Data describing other data: creation date, size, author, location, format.

method^{7,15} A function that is associated with an object and called using the dot operator; when defined inside a class it is invoked on instances of that class.

modularity^{AP} Dividing a program into independent parts each responsible for one aspect.

module^{AP 2} A file that contains Python code, including function definitions and sometimes other statements. (*AP calls this a software library.*)

modulus operator^{AP 5} An operator, %, that works on integers and returns the remainder when one number is divided by another. (*AP calls this MOD.*)

multifactor authentication^{AP} Requiring evidence from two or more categories: knowledge, possession, inherence.

multiline string⁴ A string enclosed in triple quotes that can span more than one line of a program.

N

n-gram¹² A sequence of an unspecified number of elements.

natural language¹ Any of the languages that people speak that evolved naturally.

nested conditional^{AP 5} A conditional statement that appears in one of the branches of another conditional statement.

nested list⁹ A list that is an element of another list.

newline⁵ A character that creates a line break between two parts of a string.

O

object⁸ Something a variable can refer to. An object has a type and a value.

object diagram¹⁴ A graphical representation of an object, its attributes, and their values.

object-oriented language¹⁵ A language that provides features to support object-oriented programming, notably user-defined types.

object-oriented programming¹⁴ A style of programming that uses objects to organize code and data.

open access^{AP} Research or data made available free of access restrictions.

open source^{AP} Software whose source is public and may be modified and redistributed.

open standard^{AP} A publicly available specification that anyone may implement without permission or fee.

operand¹ One of the values on which an operator operates.

operator overloading¹⁵ The process of using special methods to change the way operators work with user-defined types.

overflow error^{AP B} An error that happens when a value is too large for the number of bits available to hold it.

override¹² To replace a default value with an argument.

P

pack¹¹ Collect multiple arguments into a tuple.

packet^{AP} A chunk of data carrying metadata such as its destination, sent across a network.

packet switching^{AP} Splitting a message into packets that travel independently, possibly by different routes, and are reassembled at the destination.

pair programming^{AP} Two programmers at one workstation: one drives, one reviews and thinks ahead.

parallel computing^{AP} A program split into operations, some performed simultaneously.

parameter^{AP 3} A name used inside a function to refer to the value passed as an argument.

parent class¹⁷ A class that is inherited from.

pass^{7,A} A test whose actual result matches its expected value.

path¹³ A string that specifies a sequence of directories, often leading to a file.

pattern⁸ A rule that specifies the requirements a string has to meet to constitute a match.

persistent¹³ A persistent program runs indefinitely and keeps at least some of its data in permanent storage.

phishing^{AP} Tricking a user into revealing private information by impersonating a trusted party.

PII^{AP} Information that identifies or can be linked to a specific individual.

pixel^{AP} The smallest addressable color element of a display or image.

plagiarism^{AP} Presenting someone else's work as your own.

polymorphism¹⁶ The ability of a method or operator to work with multiple types of objects.

postcondition⁴ A requirement that should be satisfied by the function before it ends.

precondition⁴ A requirement that should be satisfied by the caller before a function starts.

procedure^{AP A} The exam's word for a named, reusable block of code, whether or not it returns a value. This book says function. In older languages the two words were distinct: a procedure returned nothing, a function returned a value.

program^{AP} A collection of statements that performs a task when run.

program function^{AP A} What a program does when it runs, described as behavior.

program input^{AP A} Data a program receives while it is running.

program output^{AP A} What a program produces: displayed text, a returned value, a file, a sound, a movement.

program purpose^{AP A} The need a program serves, or the problem it solves; why it exists.

protocol^{AP} An agreed set of rules governing system behavior.

prototype and patch¹⁴ A way of developing programs by starting with a rough draft and gradually adding features and fixing bugs.

pseudorandom^{AP 12} A pseudorandom sequence of numbers appears to be random, but is generated by a deterministic program. (*AP calls this RANDOM(a, b) — inclusive on both ends, unlike Python's range.*)

public key encryption^{AP} A public key encrypts; a different private key decrypts.

pure function^{6,14} A function that does not modify its parameters or have any effect other than returning a value.

R

reasonable time^{AP} Run time that grows polynomially or slower with input size.

receiver¹⁵ The object a method is invoked on.

recursion⁵ The process of calling the function that is currently executing.

recursive⁵ A function that calls itself is recursive.

redundancy^{AP} Duplicate paths or components that allow operation to continue after a failure.

refactoring⁴ The process of modifying a working program to improve function interfaces and other qualities of the code.

reference⁹ The association between a variable and its value.

regression^A A bug that reappears in code that used to work.

regular expression⁸ A sequence of characters that defines a search pattern.

relational operator^{AP 5} One of the operators that compares its operands: ==, !=, >, <, >=, and <=.

relative path¹³ A path that starts from the current working directory, or some other specified directory.

return value^{AP 6} The result of a function. If a function call is used as an expression, the return value is the value of the expression. (*AP calls this the RETURN statement.*)

RGB^{AP B} Color stored as three numbers, the amounts of red, green, and blue, each usually one byte.

rogue access point^{AP} An unauthorized wireless access point used to intercept network traffic.

roundoff error^{AP A,B} A loss of precision that happens because a fixed number of bits cannot represent some numbers exactly.

router^{AP} A device that forwards packets between networks toward their destination.

routing^{AP} Determining a path from sender to receiver across a network.

rubber duck debugging¹² A way of debugging by explaining a problem aloud to an inanimate object.

run-length encoding^B A lossless scheme that replaces runs of a repeated value with the value and a count.

runtime error^{AP 2} An error that causes a program to display an error message and exit.

S

sampling^{AP B} Approximating an analog signal by measuring it at regular intervals. Two independent settings: how often you measure (rate) and how precisely you record each measurement (bit depth).

scaffolding⁶ Code that is used during program development but is not part of the final version.

scalability^{AP} A system's ability to keep working acceptably as demand grows.

semantic error^{AP 2,A} An error that lets the program run but produces a wrong result. (*AP calls this a logic error.*)

sequence⁸ An ordered collection of values where each value is identified by an integer index.

sequencing^{AP} Executing steps in the order written.

sequential computing^{AP} Operations performed one at a time, in order.

serialization¹³ Converting an object to a string.

shallow copy¹⁶ A copy operation that does not copy nested objects.

shell command⁸ A statement in a shell language, which is a language used to interact with an operating system.

side effect⁶ Any effect a function has other than returning a value, such as displaying output or drawing on a canvas.

simulation^{AP} An abstraction of a real phenomenon using varying values to represent changing states, run for a purpose.

slice⁸ A part of a string specified by a range of indices.

sort key¹¹ A value, or function that computes a value, used to sort the elements of a collection.

special method¹⁵ A method that changes the way operators and some functions work with an object.

specialization¹⁷ A way of using inheritance to create a new class that is a specialized version of an existing class.

speedup^{AP} Sequential time divided by parallel time.

stack diagram³ A graphical representation of a stack of functions, their variables, and the values they refer to.

state diagram² A graphical representation of a set of variables and the values they refer to.

statement² One or more lines of code that represent a command or action.

static method¹⁵ A method that can be invoked without an object as receiver.

string^{AP 1} A type that represents sequences of characters.

string substitution⁸ Replacement of a string, or part of a string, with another string.

substring^{AP} A contiguous portion of a string.

symmetric key encryption^{AP} One shared key both encrypts and decrypts.

syntax error^{AP 1} An error in a program that makes it impossible to parse — and therefore impossible to run.

T

TCP^{AP} Transport protocol providing reliable, ordered delivery with retransmission.

test case^{AP A} A specific input paired with its expected output. Good sets include typical, boundary, and edge values.

test discovery¹⁸ A process used to find and run tests.

testing^{AP A} Checking that a program behaves correctly by running it on chosen inputs and comparing what comes out against what should have come out.

totally ordered¹⁷ A set of objects is totally ordered if we can compare any two elements and the results are consistent.

traceback³ A list of the functions that are executing, printed when an exception occurs.

traversal^{AP} Iterating over the items of a list. Full traversal visits every element; partial stops early.

trigram¹² A sequence of three elements.

Turing complete⁶ A language, or subset of a language, is Turing complete if it can perform any computation that can be described by an algorithm.

type^{AP 1} A category of values. The types we have seen so far are integers (type int), floating-point numbers (type float), and strings (type str). (*AP calls this data type.*)

U

UDP^{AP} Lightweight transport protocol with minimal error checking; faster, less reliable.

undecidable problem^{AP} A problem for which no algorithm can always give a correct yes/no answer for every input.

Unicode^{AP B} A far larger table, covering the writing systems ASCII left out.

unpack¹¹ Treat a tuple (or other sequence) as multiple arguments.

unreasonable time^{AP} Run time that grows exponentially or factorially with input size.

update⁷ An assignment statement that gives a new value to a variable that already exists, rather than creating a new variable.

V

value^{1,10} An integer, floating-point number, string, or other kind of data a program works with. In a dictionary, a value is specifically the second part of a key-value pair.

variable^{AP 2} A name that refers to a value.

W

world wide web^{AP} A system of linked pages and files that runs on top of the Internet using HTTP.

Z

zip object¹¹ The result of calling the built-in function zip, can be used to loop through a sequence of tuples.